CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 21: Trigonometrical Identities

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 The three basic identities

1. Assertion (A): If $\sin^2 A + \sin A = 1$, then $\cos^4 A + \cos^2 A = 1$. Reason (R): $1 - \sin^2 A = \cos^2 A$. (a) A true, R false (b) A false, R true (c) both true, and R is the correct reason for A (d) both true, but R is not the correct reason for A ICSE 2025, 1 mark

SOLUTION

From the given relation, $\sin A = 1 - \sin^2 A = \cos^2 A$ (using R).

Then $\cos^4 A + \cos^2 A = \sin^2 A + \sin A = 1$. So A is true, and R is exactly what proves it.

Answer: (c)

2. Statement 1: $\sin^2 \theta + \cos^2 \theta = 1$. Statement 2: $\operatorname{cosec}^2 \theta + \cot^2 \theta = 1$. Which is valid? (a) only 1 (b) only 2 (c) both (d) neither ICSE 2024, 1 mark

SOLUTION

Statement 1 is a basic identity. The correct form of the second is $\operatorname{cosec}^2 \theta - \cot^2 \theta = 1$, so statement 2 is not valid.

Answer: (a)

3. $(1 + \sin A)(1 - \sin A)$ equals: (a) $\operatorname{cosec}^2 A$ (b) $\sin^2 A$ (c) $\sec^2 A$ (d) $\cos^2 A$ ICSE 2023, 1 mark

SOLUTION

$(1 + \sin A)(1 - \sin A) = 1 - \sin^2 A = \cos^2 A$

Answer: (d)

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2 Proving identities

4. Prove that $\frac{(\cot A + \tan A - 1)(\sin A + \cos A)}{\sin^3 A + \cos^3 A} = \sec A \operatorname{cosec} A$. ICSE 2025, 4 marks

SOLUTION

$\cot A + \tan A - 1 = \frac{\cos^2 A + \sin^2 A - \sin A \cos A}{\sin A \cos A} = \frac{1 - \sin A \cos A}{\sin A \cos A}$

$\sin^3 A + \cos^3 A = (\sin A + \cos A)(\sin^2 A - \sin A \cos A + \cos^2 A) = (\sin A + \cos A)(1 - \sin A \cos A)$

$\text{LHS} = \frac{(1 - \sin A \cos A)(\sin A + \cos A)}{\sin A \cos A\,(\sin A + \cos A)(1 - \sin A \cos A)} = \frac{1}{\sin A \cos A}$

$= \sec A \operatorname{cosec} A$ = RHS. **Proved.**

5. Factorise $\sin^3 \theta + \cos^3 \theta$. Hence prove that $\frac{\sin^3 \theta + \cos^3 \theta}{\sin \theta + \cos \theta} + \sin \theta \cos \theta = 1$. ICSE 2024, 4 marks

SOLUTION

Using $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$: $\sin^3 \theta + \cos^3 \theta = (\sin \theta + \cos \theta)(\sin^2 \theta - \sin \theta \cos \theta + \cos^2 \theta)$

$= (\sin \theta + \cos \theta)(1 - \sin \theta \cos \theta)$

$\text{LHS} = \frac{(\sin \theta + \cos \theta)(1 - \sin \theta \cos \theta)}{\sin \theta + \cos \theta} + \sin \theta \cos \theta = 1 - \sin \theta \cos \theta + \sin \theta \cos \theta$

$= 1$ = RHS. **Proved.**

6. Prove that $(\sin^2 \theta - 1)(\tan^2 \theta + 1) + 1 = 0$. ICSE 2023, 4 marks

SOLUTION

$\sin^2 \theta - 1 = -\cos^2 \theta$ and $\tan^2 \theta + 1 = \sec^2 \theta$

$\text{LHS} = -\cos^2 \theta \times \sec^2 \theta + 1 = -1 + 1$

$= 0$ = RHS. **Proved.**

7. Prove that $\frac{\sin A}{1 + \cot A} - \frac{\cos A}{1 + \tan A} = \sin A - \cos A$. ICSE 2020, 3 marks

SOLUTION

$\frac{\sin A}{1 + \cot A} = \frac{\sin A}{\frac{\sin A + \cos A}{\sin A}} = \frac{\sin^2 A}{\sin A + \cos A}$ and $\frac{\cos A}{1 + \tan A} = \frac{\cos^2 A}{\sin A + \cos A}$

$\text{LHS} = \frac{\sin^2 A - \cos^2 A}{\sin A + \cos A} = \frac{(\sin A - \cos A)(\sin A + \cos A)}{\sin A + \cos A}$

$= \sin A - \cos A$ = RHS. **Proved.**

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8. Prove that $\left(\frac{1-\tan\theta}{1-\cot\theta}\right)^2 = \tan^2\theta$. ICSE 2020, 4 marks

SOLUTION

$1-\cot\theta = 1-\frac{1}{\tan\theta} = \frac{\tan\theta-1}{\tan\theta}$

$\frac{1-\tan\theta}{1-\cot\theta} = \frac{(1-\tan\theta)\tan\theta}{\tan\theta-1} = -\tan\theta$

Squaring: LHS $= (-\tan\theta)^2 = \tan^2\theta$ = RHS. **Proved.**

9. Prove that $(\operatorname{cosec}\theta-\sin\theta)(\sec\theta-\cos\theta)(\tan\theta+\cot\theta)=1$. ICSE 2019, 3 marks

SOLUTION

$\operatorname{cosec}\theta-\sin\theta = \frac{1-\sin^2\theta}{\sin\theta} = \frac{\cos^2\theta}{\sin\theta}$ and $\sec\theta-\cos\theta = \frac{\sin^2\theta}{\cos\theta}$

$\tan\theta+\cot\theta = \frac{\sin^2\theta+\cos^2\theta}{\sin\theta\cos\theta} = \frac{1}{\sin\theta\cos\theta}$

LHS $= \frac{\cos^2\theta}{\sin\theta}\times\frac{\sin^2\theta}{\cos\theta}\times\frac{1}{\sin\theta\cos\theta} = 1$ = RHS. **Proved.**

10. Prove that $\sqrt{\sec^2\theta+\operatorname{cosec}^2\theta} = \tan\theta+\cot\theta$. ICSE 2018, 3 marks

SOLUTION

$\sec^2\theta+\operatorname{cosec}^2\theta = \frac{1}{\cos^2\theta}+\frac{1}{\sin^2\theta} = \frac{\sin^2\theta+\cos^2\theta}{\sin^2\theta\cos^2\theta} = \frac{1}{\sin^2\theta\cos^2\theta}$

LHS $= \frac{1}{\sin\theta\cos\theta}$

RHS $= \frac{\sin\theta}{\cos\theta}+\frac{\cos\theta}{\sin\theta} = \frac{\sin^2\theta+\cos^2\theta}{\sin\theta\cos\theta} = \frac{1}{\sin\theta\cos\theta}$

LHS = RHS. **Proved.**

11. Prove that $(1+\cot\theta-\operatorname{cosec}\theta)(1+\tan\theta+\sec\theta)=2$. ICSE 2018, 4 marks

SOLUTION

LHS $= \left(\frac{\sin\theta+\cos\theta-1}{\sin\theta}\right)\left(\frac{\cos\theta+\sin\theta+1}{\cos\theta}\right) = \frac{(\sin\theta+\cos\theta)^2-1}{\sin\theta\cos\theta}$

$= \frac{1+2\sin\theta\cos\theta-1}{\sin\theta\cos\theta} = 2$ = RHS. **Proved.**

12. Prove that $\frac{\sin\theta-2\sin^3\theta}{2\cos^3\theta-\cos\theta} = \tan\theta$. ICSE 2017, 3 marks

SOLUTION

LHS $= \frac{\sin\theta(1-2\sin^2\theta)}{\cos\theta(2\cos^2\theta-1)}$

Using $1=\sin^2\theta+\cos^2\theta$: $1-2\sin^2\theta = \cos^2\theta-\sin^2\theta$ and $2\cos^2\theta-1 = \cos^2\theta-\sin^2\theta$.

The brackets cancel: LHS $= \frac{\sin\theta}{\cos\theta} = \tan\theta$ = RHS. **Proved.**

3 Complementary angles

13. Without using tables, evaluate $\sin^2 28^\circ + \sin^2 62^\circ + \tan^2 38^\circ - \cot^2 52^\circ + \frac{1}{4}\sec^2 30^\circ$. ICSE 2017, 3 marks

SOLUTION

$\sin 62^\circ = \cos 28^\circ$, so $\sin^2 28^\circ + \sin^2 62^\circ = \sin^2 28^\circ + \cos^2 28^\circ = 1$.

$\cot 52^\circ = \tan 38^\circ$, so $\tan^2 38^\circ - \cot^2 52^\circ = 0$.

$\frac{1}{4}\sec^2 30^\circ = \frac{1}{4} \times \frac{4}{3} = \frac{1}{3}$

Answer: $1 + 0 + \frac{1}{3} = \frac{4}{3}$

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